Cognitive Diagnosis Focusing on Knowledge Concepts

Sheng Li
lsjnu@stu2020.jnu.edu.cn
College of Information Science and Technology, Jinan University
Guangdong Institution of Smart Education, Jinan University
Guangzhou, Guangdong, China

Quanlong Guan*
gql@jnu.edu.cn
College of Information Science and Technology, Jinan University
Guangdong Institution of Smart Education, Jinan University
Guangzhou, Guangdong, China

Liangda Fang*
fangld@jnu.edu.cn
College of Information Science and Technology, Jinan University
Pazhou Lab
Guangzhou, Guangdong, China

Fang Xiao
xf990616@stu2021.jnu.edu.cn
College of Information Science and Technology, Jinan University
Guangdong Institution of Smart Education, Jinan University
Guangzhou, Guangdong, China

Zhenyu He
tzhenyuhe@jnu.edu.cn
College of Information Science and Technology, Jinan University
Guangdong Institution of Smart Education, Jinan University
Guangzhou, Guangdong, China

Yizhou He
hyz@jnu.edu.cn
College of Management, Jinan University
Guangdong Institution of Smart Education, Jinan University
Guangzhou, Guangdong, China

Weiqi Luo
lwq@jnu.edu.cn
College of Information Science and Technology, Jinan University
Guangdong Institution of Smart Education, Jinan University
Guangzhou, Guangdong, China

ABSTRACT

Cognitive diagnosis is a crucial task in the field of educational measurement and psychology, which aims to diagnose the strengths and weaknesses of participants. Existing cognitive diagnosis methods only consider which of knowledge concepts are involved in the knowledge components of exercises, but ignore the fact that different knowledge concepts have different effects on practice scores in actual learning situations. Therefore, researchers need to reshape the learning scene by combining the multi-factor relationships between knowledge components. In this paper, in order to more comprehensively simulate the interaction between students and exercises, we developed a neural network-based CDMFKC model for cognitive diagnosis. Our method not only captures the nonlinear interaction between exercise characteristics, student performance, and their mastery of each knowledge concept, but also further considers the impact of knowledge concepts by designing the difficulty and discrimination of knowledge concepts, and uses multiple neural layers to model their interaction so as to obtain accurate and interpretable diagnostic results. In addition, we propose an improved CDMFKC model with guessing parameter and slipping parameter designed by knowledge concept proficiency and student proficiency vectors. We validate the performance of these two diagnostic models on six real datasets. The experimental results show that the two models have better effects in the aspects of accuracy, rationality and interpretability.

CCS CONCEPTS

• **Computing methodologies → Knowledge representation and reasoning**.

KEYWORDS

Cognitive Diagnosis, Neural Network, Knowledge Concepts, Guessing Parameter, Slipping Parameter

ACM Reference Format:
Sheng Li, Quanlong Guan, Liangda Fang, Fang Xiao, Zhenyu He, Yizhou He, and Weiqi Luo. 2022. Cognitive Diagnosis Focusing on Knowledge Concepts. In *Proceedings of the 31st ACM International Conference on Information and Knowledge Management (CIKM '22), October 17–21, 2022, Atlanta, GA, USA.* ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/3511808.3557096

1 INTRODUCTION

In recent years, the rapid development of related information technologies such as artificial intelligence and big data has brought new challenges and opportunities to various industries [3]. In the field of education, most of the traditional evaluation methods are manual evaluation. Manual evaluation is time-consuming,

*Both authors are corresponding authors.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
CIKM '22, October 17–21, 2022, Atlanta, GA, USA
© 2022 Association for Computing Machinery.
ACM ISBN 978-1-4503-9236-5/22/10…$15.00
https://doi.org/10.1145/3511808.3557096

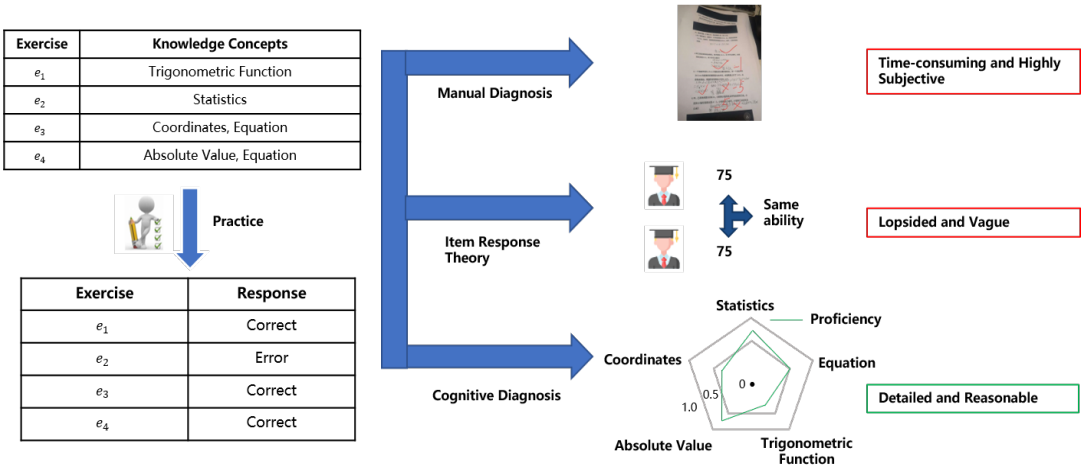

Figure 1: Example diagram of cognitive diagnosis and traditional evaluation. Students get scores by choosing some exercises to answer. The cognitive diagnosis process was used to diagnose the degree of students'mastery of each knowledge concept.

labor-intensive and subjectivity [1]. Under this context, cognitive diagnostic theory came into being [18].

Cognitive diagnosis aims to discover the states of students in the learning process, such as students' specific mastery of each knowledge concept [22]. Figure 1 shows an example of a cognitive diagnosis process compared to traditional methods. Cognitive diagnosis obtains the students' mastery degree of each knowledge concept (*e.g.*, Statistics) based on the correctness of students' answer on some exercise questions. The diagnosis results have many practical applications, such as exercise recommendation [14, 27] and adaptive learning [4, 21]. Challenges of improving the accuracy and applicability of cognitive diagnostics involves rational modeling among student scores, knowledge composition, and exercise factors [10].

Many scholars have conducted a lot of research on cognitive diagnosis, such as Deterministic Inputs, Noise And gate model (DINA) [13] and its variants [6–9], Item Response Theory (IRT) [23] and its multidimensional extension (MIRT) [25] and Matrix Factorization (MF) [17]. These models have achieved some good results, but still use hand-designed interaction functions. They are a linear combination of the features of students and exercises, mostly of which are solved by parameter estimation. Hence, it cannot capture the complex connection between students and exercises.

To address this defect, some scholars designed a neural network-based cognitive diagnosis framework [31], which incorporates neural networks to learn the complex exercising interactions, for obtaining both accurate and interpretable diagnosis results. However, we observe that these models only consider which of knowledge concepts are involved in the exercise. In actual learning situations, knowledge concepts in different knowledge components have different effects on exercise scores. For example, exercise j contains the knowledge concepts "multiplication" and "addition". Student A's proficiencies in multiplication and addition are 0.8 and 0.6, respectively while student B's proficiencies in them are 0.8 and 0.6, respectively. In the previous cognitive diagnosis methods, because their knowledge concept proficiency is greater than the knowledge concept difficulty, students A and B are considered to have the same probability of scoring. It is considered here that multiplication and addition have the same effect on exercise j. Indeed, the knowledge concept multiplication has a greater impact on the practice score than addition. Hence, student A is more likely to answer the exercise correctly. It is necessary to reshape the learning situation by combining the multi-factor relationships between knowledge concepts.

In this paper, we propose Cognitive Diagnostic Model Focusing on Knowledge Concept (CDMFKC). CDMFKC considers the effects of different knowledge concepts on exercise scores, and utilizes neural network to model the complex interaction between students and exercise attributes. We also consider the influence of the impact of knowledge concepts in answer prediction, and dynamically adjust the judgment boundary according to the difference of students and exercise, which makes the model more applicable and more accurate. Our key contributions are listed as follows.

- We propose the notion of impacts of different knowledge concepts in the knowledge component on student scores.
- We present the CDMFKC model and conduct comparative experiments on six real-world datasets with some classical models to verify the accuracy, applicability, and interpretability of our model.
- We design guessing and slipping parameters, and propose GSCDMFKC to fit the real learning scene to achieve more accurate diagnosis. The rationality of the model and the effects of each attribute are verified by ablation experiments.

The rest of our paper is organized as follows. We present the related work in Section 2, followed by a description of the basic design work in Section 3. After introducing our model in detail in Section 4, we present experimental results on real datasets in Section 5, and finally draw conclusions in Section 6.

2 RELATED WORK

In this section, we will review the previous research work from three aspects.

Classical Cognitive Diagnosis. The research on cognitive diagnostic model mainly comes from the field of educational psychology, and it has become a research hotspot. More than 60 cognitive diagnostic models have been developed [14], of which DINA [13] and IRT [23] are the most widely used and studied. DINA model uses Q-matrix theory, which is usually calibrated by experts. As shown in Figure 4, Exercise 1 examines the knowledge concepts "Trigonometric Function" and "Logarithm versus Exponential", so it is marked as 1. The remaining two knowledge concepts are not examined, so it is marked as 0. DINA describes the exercise as a vector containing one or more knowledge concepts, and introduces sliding parameters and guessing parameters to model the students' cognitive state. As a classic non-compensatory model, DINA stipulates that student i can only correctly answer exercise j if he has mastered all the knowledge concepts required by exercise j. Let n_{ij} denote the potential answering situation of student i on exercise j with K knowledge concepts, which is defined as:

$$n_{ij} = \prod_{k=1}^{K} \alpha_{ik}{}^{q_{jk}} \tag{1}$$

where $n_{ij} = 1$ represents that student i correctly answers exercise j and $n_{ij} = 0$ otherwise; $q_{jk} = 1$ means that exercise j involves knowledge concept k and $q_{jk} = 0$ otherwise; $\alpha_{ik} = 1$ means that student i masters knowledge concept k and $\alpha_{ik} = 0$ otherwise. After introducing two exercise factors (guessing and slipping), the probability of student i answering exercise j correctly is modeled as:

$$p(r_{ij}|\alpha_i) = g_j^{1-n_{ij}}(1-s_j)^{n_{ij}} \tag{2}$$

where r_{ij} represents the score of student i on exercise j, α_i refers to students' mastery of all knowledge concepts, g_j is the guessing parameter of exercise j, and s_j is the slipping parameter.

On the other hand, in IRT the probability of student i answering exercise j correctly is modeled as:

$$p(r_{ij}|\alpha_i) = c_j + \frac{1-c_j}{1+e^{-da_j(\alpha_j-b_j)}}, \tag{3}$$

where a_j denotes the discrimination of exercise j, b_j indicates the difficulty of exercise j and c_j indicates the coefficient of guess for exercise j. However, these models rely on manually designed interaction functions, which are time-consuming and labor-intensive, limiting their further applications.

Data Mining Methods. Matrix factorization (MF) has been proved to be an effective application in cognitive diagnosis [11]. Singular value decomposition (SVD) factorizes the score matrix and obtains latent eigenvectors for students and exercises [29]. The weighted and multi-relational factor decomposition model to profile students with the main relationships of the target variables [28]. A probabilistic matrix factorization framework is based on two classical educational theories and an educational priority tracking students' knowledge level [5]. However, when applied to the field of cognitive diagnosis, these methods have poor interpretability, because there is no clear correspondence between the elements in the feature vector and specific knowledge concepts.

Neural Networks. With the rapid development of neural network (NN), breakthrough achievements have been made in many fields, such as pattern recognition [16], computer vision [30], adaptive system [15], intelligent robot [19]. Neural networks also have some research applications in education, such as problem difficulty prediction [2] and student performance prediction [20]. The most interesting applications of cognitive diagnosis are deep knowledge tracking (DKT) [24] and NeuralCDM (NCDM) [31]. Deep Knowledge Tracking is the first attempt to predict student scores using recurrent neural networks (RNNs), but does not take into account the complex interactions between students and exercise. NCDM uses neural networks to model learning interactions and applies monotonicity assumptions to ensure model interpretability. The monotonicity assumption means that in any dimension of students' knowledge level, the probability of students answering the exercises correctly is monotonically increasing with the increase of knowledge proficiency. However, NCDM does not consider the impact of different knowledge concepts on the score, which is an important issue in real teaching scenarios.

Table 1: The notations used in this paper.

Notations	Descriptions
ψ	Tuple representation of a learning system,$\psi = (S, E, K, R, Q, KI)$
S	A set of students, s_i represents the ith student
E	A set of exercises, e_j represents the jth exercise
K	A set of knowledge concepts, k_i represents the ith knowledge concept
R	A Student Response Record Triple, $R = (s, e, r)$
Q	Q-matrix, reflecting the relationship between exercises and knowledge concepts
KI	Impact of Knowledge Concepts on Practice Answering, $KI = \{k_1^{imp}, k_2^{imp}, \cdots, k_M^{imp}\}$
V_i	Student s_i's proficiency vector of knowledge concepts
y_i'	Student s_i's predicted response score
y_i	Student s_i's true answer score
w_{ij}'	Student s_i's dynamic judgment boundary for exercise e_j
G_{ij}	Student s_i's guessing parameters for exercise e_j
S_{ij}	Student s_i's slipping parameters for exercise e_j
h_j^{diff}	Knowledge concept difficulty vector for exercise e_j
h_j^{disc}	Discrimination value for exercise e_j
h_j^{imp}	Knowledge concept impact of exercise e_j
g_j^{imp}	Knowledge concept impact of exercise e_j

3 BASIC DESIGN WORKS

This section will introduce our problem definition and design. Table 1 summarizes the symbols used in each module in this paper.

3.1 Problem Description

Definition 3.1 (Learning system). Suppose there are N students, M practice questions and K knowledge concepts. A learning system ψ can be represented as a tuple (S, E, K, R, Q, KI), where

- $S = \{s_1, s_2, \cdots, s_N\}$ represents the set of N students.
- $E = \{e_1, e_2, \cdots, e_M\}$ represents the set of M exercises.
- $K = \{k_1, k_2, \cdots, k_K\}$ represents the set of K knowledge concepts.
- $R = (s, e, r)$ is a set of triples, representing the record of the exercise, where $s \in S$, $e \in E$ and $r \in \{0, 1\}$ represents the score of the student s on the exercise e. Where r = 1 when student s answers exercise e correctly otherwise r = 0.
- $Q = \{Q_{ij}\}_{M\times K}$ is the Q-matrix, which represents the connection between exercises and knowledge concepts. $Q_{ij} = 1$ means that exercise e_i examines the knowledge concept k_j otherwise $Q_{ij} = 0$.

Table 2: Cognitive Diagnostic Model Factor Inclusion Comparison. Records: Exercise records. Knowledge: Knowledge composition. Diff & Disc: Knowledge difficulty and Exercise discrimination. Impact: Impact of knowledge. Guess & Slip: Guessing and Slipping Parameters.

	PMF	IRT & MIRT	DINA	NCDM	CDMFKC	GSCDMFKC
Records	Y	Y	Y	Y	Y	Y
Knowledge	Y	Y	Y	Y	Y	Y
Diff & Disc		Y		Y	Y	Y
Impact					Y	Y
Guess & Slip			Y			Y

- $KI = \{k_1^{imp}, k_2^{imp}, \cdots, k_M^{imp}\}$ represents the impact of knowledge concepts contained in M exercises. For example, $k_j^{imp} = [0.3, 0.5, 0.4]$ means that exercise j contains three knowledge concepts, and the second knowledge concept has a greater impact on the exercise than the other ones.

Definition 3.2 (Cognitive diagnosis). Given a learning system $\psi = (S, E, K, R, Q, KI)$, the task of cognitive diagnosis is to obtain students' knowledge concepts proficiency.

3.2 Elements Required for the System

In this section, we introduce some of the factors that our cognitive diagnostic system focuses on and compare with traditional models shown in Table 2.

Exercises Records. In a learning system, students' exercises records are very important. With students' exercises records, we can analyze students' knowledge proficiency and predict answers.

Knowledge Components. The knowledge component provides the link between exercises and knowledge concepts. For example, the knowledge component of exercise e_3 in Figure 1 contains two knowledge concepts: coordinate and equation. In this paper, we make full use of knowledge components so as to improve the accuracy of cognitive diagnosis.

Knowledge Difficulty and Exercise Discrimination. Knowledge difficulty h^{diff} and exercise discrimination h^{disc} are based on IRT [23]. The former h^{diff} indicates the difficulty of knowledge concepts in the exercise while the latter h^{disc} is the ability of the exercises to distinguish students with different levels of knowledge. Considering both parameters makes our diagnosis more reasonable. We will elaborate it in detail in Chapter 4.

Guessing and Slipping Parameters. In a learning scenario, students who do not master all the knowledge concepts may also answer correctly because of guessing. Similarly, students who have mastered all knowledge concepts may also answer incorrectly due to slipping. Hence, based on IRT, we introduce guessing and slipping parameters to fit the real scene.

Impact of Knowledge Concepts. The impact of knowledge concepts represents the different effects of knowledge concepts on exercise score. In a learning scenario, different knowledge concepts have different impacts. For example, in an exercise that includes the concepts of "multiplication" and "addition", "multiplication" has a greater impact on exercise score. So in order to achieve a more accurate diagnosis result, we need to consider the impact of different knowledge concepts in the exercise on the exercise score, which will be introduced in detail in Section 4.

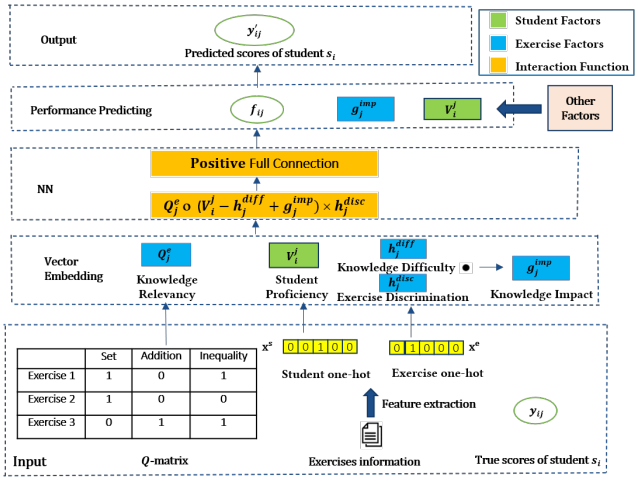

Figure 2: Description of CDMFKC Model Diagnosis Process.

Our CDMFKC takes into account the impact of knowledge concepts parameter compared with other models, and further simulates ehe influence of guessing parameters and slipping parameters, shown in Table 2.

4 MODEL DESCRIPTION

In this section, we first introduce the definition of the impact of knowledge concepts, then elaborate the detailed design of the CDMFKC model and finally present an improved GSCDMFKC model.

4.1 Impact of Knowledge Concept

Most previous cognitive diagnostic models use the Q-matrix to represent the link between exercise and knowledge concepts. However, only relying on the Q-matrix cannot distinguish the impacts of different knowledge concepts on exercise scores. For example, there are 4 knowledge concepts $\{k_1, k_2, k_3, k_4\}$ and the Q-matrix of exercise e is $[1, 0, 0, 1]$, which means that exercise e involves knowledge concepts k_1 and k_4. Although we can obtain the difficulty of exercise e via diagnosis, it only reflects the degree of difficulty that students answer exercise e under the assumption that the two knowledge concepts k_1 and k_4 have the same impacts on the exercise. However, the impact of different knowledge concepts are in general different in actual scenarios. For example, the impact of the knowledge concepts "multiplication" and "addition" in a multiplication exercise is different. We observed that a metric is needed to measure the impact of different knowledge concepts on exercise scores.

Knowledge Concept Impact. Suppose that there are N students, M exercises and K knowledge concepts. Our purpose is to obtain the impact of K knowledge concepts on different exercises, which is a a $1 \times K$ dimensional vector. As illustrated in Introduction, exercise j contains the knowledge concepts "multiplication" and "addition". Student A's proficiencies in multiplication and addition are 0.8 and 0.6, respectively while student B's proficiencies in them are 0.8 and 0.6, respectively. Student A is more likely to answer

the exercise correctly. We define the impact of knowledge concepts in exercise j as $h_j^{imp} \in (0,1)^{1\times K}$, which indicates the different impacts of K knowledge concepts of exercise j on exercise scores. The formula is expressed as follows:

$$h_j^{imp} = \sigma(x_j^e \times M), \tag{4}$$

where x_j^e is the one-hot vector representation of exercise j, M is a trainable matrix and σ is the *Sigmod* activation function.

In the following, we discuss the link between other exercise factors and the impact of knowledge concepts. CDMFKC utilizes neural networks to simulate the complex interaction of exercise factors and student factors. In CDMFKC, the discrimination degree of exercise j is expressed as $h_j^{disc} \in (0,1)$, which represents the ability of exercise j to distinguish students with different knowledge mastery levels. Based on exercise difficulty, we design the knowledge concept difficulty $h_j^{diff} \in (0,1)^{1\times K}$ contained in exercise j as a $1 \times K$ dimension matrix, which represents the difficulty of students mastering K knowledge concepts.

The Q-matrix establishes the connection between the exercise and the knowledge concept, and both the exercise discrimination h_j^{disc} and the knowledge concept difficulty h_j^{diff} are related to the knowledge concept. Therefore, we use the joint knowledge concept impact g_j^{imp} to consider the impact of discrimination and knowledge concept difficulty, which can be expressed as follows:

$$g_j^{imp} = \sigma(\alpha \cdot h_j^{imp} + \beta \cdot h_j^{diff} \cdot h_j^{disc}), \tag{5}$$

where α and β are hyperparameters. We will discuss the effect of h_j^{imp} and g_j^{imp} in Section 5.3.

Monotonicity assumption. We observed that in IRT, the monotonicity assumption was used to guarantee the interpretability of the exercise and student factors [25]. The monotonicity assumption means that the probability of correct response to the exercise is monotonically increasing at any dimension of the student's knowledge proficiency. Similarly, it is assumed that in any dimension of knowledge concept, the effect of knowledge concept on exercise score is monotonically increasing as the impact of knowledge concept increases.

4.2 CDMFKC Model

Given a learning system $\psi = (S, E, K, R, Q, KI)$, we can extract the one-hot vector of students representing $x^s \in (0,1)^{1\times N}$ and the one-hot vector of exercises representing $x^e \in (0,1)^{1\times M}$. Mathematically, we formulate the output of the CDMFKC model as:

$$y'_{ij} = PP(NN(VE(IN(x^s, x^e), Q))) \tag{6}$$

where y'_{ij} indicates the predicted score of student i in exercise j; IN, VE, NN and PP are input layer, vector embedding layer, neural network layer and performance prediction layer respectively. We will describe each layer in detail below.

Input Layer. In this layer, the information we include the Q-matrix that reflects the connection between exercises and knowledge concepts, the information e of the exercise and the students' real score y_{ij} on the exercise. When we train, the batch parameter of each input training is 32. After we read the numbered sets U_e and U_s of exercises and students in a batch, they are converted into one-hot vectors $x^s \in (0,1)^{1\times N}$ and $x^e \in (0,1)^{1\times M}$. For example, x_i^s represents the i-th student and x_j^e represents the j-th exercise.

Vector Embedding Layer. In this layer, we design the vector of our embedded layer based on the information provided by the input layer.

- We represent each student with a skill proficiency vector, which is also the goal of our final diagnosis. The skill proficiency vector of student s_i can be expressed as:

$$V_i = \sigma(x_i^s \times A), \tag{7}$$

  where $V_i \in (0,1)^{1\times K}$, $A \in \mathbb{R}^{N\times K}$ is a trainable matrix and σ is an activation function which we use *sigmod* in this paper where

$$\sigma(x) = \frac{1}{1+e^{-x}}, \tag{8}$$

- The relationship between exercises and knowledge concepts is represented by Q-matrix, which is Q_j^e here, the formula is as follows:

$$Q_j^e = x_j^e \times Q, \tag{9}$$

  where $Q_j^e \in (0,1)$ and Q is a predetermined Q-matrix.
- Knowledge difficulty $h_j^{diff} \in (0,1)^{1\times K}$ represents the different difficulty of each knowledge concept examined by exercise e_j, and exercise discrimination $h_j^{disc} \in (0,1)$ represents the ability of exercise e_j to distinguish students with different skill proficiency levels. Both are extended from IRT, and the formulas are:

$$h_j^{diff} = \sigma(x_j^e \times B), \tag{10}$$

$$h_j^{disc} = \sigma(x_j^e \times C), \tag{11}$$

  where $B \in \mathbb{R}^{M\times K}$, $C \in \mathbb{R}^{M\times 1}$ are trainable matrices and σ is an activation function.
- The influence of knowledge concept g_j^{imp} is shown in formula 5.

Neural Network Layer. Inspired by the design of MIRT model, we connect the parameters of vector embedding layer to get our input as follows:

$$x_{ij} = Q_j^e \circ (V_i - h_j^{diff} + g_j^{imp}) \times h_j^{disc}, \tag{12}$$

where $\circ$ is element-wise product.

Then the design of our full connection layer is as follows:

$$f_{ij} = \Phi(W \times x_{ij} + b), \tag{13}$$

where f_{ij} refers to the probability that students s_i correctly answer exercise e_j, Φ is an nonlinear interaction function (*sigmod* used in this paper), and W and B are trainable parameters. The number of network layers can be selected within an appropriate range. We use two full connection layers and one output layer. To satisfy the monotonicity assumption, we use a method to restrict the element W of each layer of the network to be positive.

Performance Prediction Layer. In this layer, we consider the dynamic influence of some factors on the response probability. After training in the neural network layer, we can obtain the proficiency vector V_i^j of students s_i for the knowledge concepts contained in exercise e_j, which is the main purpose of our cognitive

diagnosis. f_{ij} represents the predicted score of student s_i's response to exercise e_j after training. However, we noticed that the previous methods (such as NCDM) used 0.5 as the limit of right and wrong judgment, which is too absolute in the actual teaching scene. We observed that the impact of knowledge concepts indicates the magnitude of the impact of different knowledge concepts on exercise responses. The higher the impact of the knowledge concepts contained in exercise e_j, the higher the requirements for the proficiency of students' knowledge concepts. For example, exercises A and B contain the same knowledge concept K, and the impact of the knowledge concept h_A^{imp} is greater than h_B^{imp}, then the score requirement of exercise A is higher than that of exercise B. The method we adopted was to compare the average knowledge concept impact AVG_j^{imp} of the knowledge concept examined in exercise e_j with the average knowledge concept impact AVG_{ij}^{imp} of all exercises done by student s_i. Their formulas are as follows:

$$AVG_j^{imp} = \frac{1}{K_j}\sum_{k=1}^{K_j} g_k^{imp}, \tag{14}$$

$$AVG_{ij}^{imp} = \frac{1}{C}\sum_{j=1}^{M_i}\sum_{k=1}^{K_j} g_k^{imp}, \tag{15}$$

where k is the number of knowledge concepts examined in exercise e_j, g_k^{imp} is the impact of knowledge concepts of knowledge concept K and $C = M_i \times K_j$, M_i is the number of exercises done by student s_i.

We compare AVG_j^{imp} and AVG_{ij}^{imp} to determine whether to adjust the judgment limit to achieve the purpose of dynamic diagnosis of different students. The correction parameter w_{ij} is expressed by the formula as follows:

$$w_{ij} = AVG_j^{imp} \times \delta(AVG_{ij}^{imp}, AVG_j^{imp}), \tag{16}$$

where δ is the indicator function, $\delta(x, y) = 1$ if $x > y$ otherwise $\delta(x, y) = 0$. Further, we adjust the judgment boundary of prediction according to w_{ij}, which also simulates the fact that knowledge concepts have different influences in the real teaching process. For example, when the average knowledge concept impact AVG_j^{imp} of exercise e_j is lower than the average knowledge concept impact AVG_{ij}^{imp} of all exercises done by student s_i, exercise e_j requires less knowledge concept proficiency of students, and the judgment boundary w'_{ij} is adjusted to $0.5 - w_{ij}$. The experimental results in Section 5 demonstrate its effectiveness.

Output Layer. Finally, the output layer gets the adjusted student's prediction and answers y'_{ij}, We can do some analysis at this level. In this paper, the cross-entropy between the true score y_{ij} and the predicted score y'_{ij} is used to design loss function in CDMFKC model training, which is defined as follows:

$$loss = -\sum_i \sum_j (y_{ij} \log y'_{ij} + (1 - y_{ij} \log(1 - y'_{ij}))), \tag{17}$$

4.3 GSCDMFKC Model

As mentioned in Section 2, some students still answer correctly by guessing and answer incorrectly due to mistakes. Therefore, it is reasonable for us to introduce guess parameters and slip parameters into Other Factors module shown in Figure 2 of the performance prediction layer, which we name as GSCDMFKC model.

After training, we get the predicted output f_{ij}. When $f_{ij} > 0.5$, we assume that the students will answer correctly, otherwise they will answer incorrectly. Then, when predicting that students can answer correctly, from the monotonicity assumption, it can be obtained that the average proficiency $AVG(h^s)$ of student s_i for the knowledge concepts examined by exercise e_j is greater than the average knowledge concept difficulty $AVG(h^{diff})$. The slipping parameter S_{ij} formula is expressed as follows:

$$S_{ij} = AVG_j^{imp} \frac{\sum_{k=1}^{K_j} h_{ijk}^{diff}}{\sum_{k=1}^{K_j} h_{ijk}^{s}}, \tag{18}$$

where h_{ijk}^{diff} is the knowledge concept difficulty of student s_i on the knowledge concept k examined by exercise e_j, and h_{ijk}^{s} is the knowledge concept proficiency.

Similarly, when predicting that students will answer incorrectly, we consider the guessing parameter G_{ij}, which is as follows:

$$G_{ij} = AVG_j^{imp} \frac{\sum_{k=1}^{K_j} h_{ijk}^{s}}{\sum_{k=1}^{K_j} h_{ijk}^{diff}}, \tag{19}$$

Then we can get the adjusted predicted answer probability y'_{ij}, the formula is as follows:

$$y'_{ij} = (1 - q_{ij}) \times f_{ij} \tag{20}$$

where $q_{ij} = S_{ij}$ when the students are predicted to answer the exercise correctly, otherwise $q_{ij} = G_{ij}$. Finally, compare y'_{ij} with the judgment condition w'_{ij} adjusted by the average knowledge concept impact correction parameter w_{ij} to obtain the predicted score. When $y_{ij} = 1$ and $y'_{ij} > \omega_{ij}$ or $y_{ij} = 0$ and $y'_{ij} < \omega_{ij}$, it is predicted that student s_i can answer exercise e_j correctly, otherwise the student will be judged to answer incorrectly.

Since each student's knowledge proficiency is different, our purpose of dynamically adjusting the answer probability is achieved. From the known students' real answers y_{ij}, we can get whether the prediction of our model is correct and calculate the prediction accuracy. We make full use of each element in the learning system, and the experimental results in Section 5 also show that our optimization is beneficial to improve the prediction accuracy.

5 EXPERIMENT

To verify the validity and interpretability of the proposed model in performance prediction tasks, we performed extensive experiments on six real data sets and compared them with some baselines. In this section, we first briefly introduce the dataset, then analyze and discuss the experimental results.

5.1 Overview of Datasets

Our experiment used six real-world datasets, ASSIST [33], Math1, and Math2 [32], Algebra1, Algebra2, and Bridge to Algebra [26].

To guarantee that each student to have enough data to diagnose, we selected students with at least 15 records. In addition, the data is divided and 70% is selected as the training set, 10% as the validation set, and 20% as the test set. We use the training set to train the

Table 3: Dataset summary.

Dataset	ASSIST	Math1	Math2	Algebra1	Algebra2	Bridge
#Students	4,163	4,209	3,911	574	1,338	1,146
#Exercises	17,746	20	20	1,084	91,913	19,355
#Knowledge concepts	123	11	16	113	492	494
#Response logs	324,572	14,246	19,555	809,694	2,270,384	3,679,199
#Knowledge concepts per exercise	1.19	3.25	3.2	8.40	5.59	4.99
AVG_{res}	8.05	7.3	3.36	7.13	1.93	14.33
$STD_{res>1}$	0.316	0.304	0.325	0.322	0.313	0.332

Table 4: Comparison of knowledge concept impact results. h_j^{imp} means that the influence of knowledge concept difficulty and exercise discrimination is not considered, while g_j^{imp} considers both.

Dataset	Accuracy	
	h_j^{imp}	g_j^{imp}
ASSIST	0.739 ± 0.002	**0.762 ± 0.001**
Math1	0.728 ± 0.003	**0.740 ± 0.003**
Math2	0.714 ± 0.001	**0.732 ± 0.002**

model, use the validation set to verify the quality of the model and adjust the parameters, and verify the test set to test the model.

The students' math knowledge level is stable during the exam, however, the knowledge proficiency of students varies with time. In order to analyze the stability of students' knowledge state in ASSIST dataset, we select two indicators to evaluate. The first is the average number of logs per student for each concept of knowledge, as shown in the following formula:

$$AVG_{res} = \frac{\sum_i^N \sum_j^K res(i,j)}{\sum_i^N \sum_j^K I(res(i,j) > 0)}, \tag{21}$$

where $res(i,j)$ represents the exercise containing knowledge concept k_j answered by student i, I is the indicator function. Another indicator is the standard deviation of score r_{ij} when $res(i,j) > 1$, as shown in the following formula:

$$STD_{res>1} = mean(mean(Std_{ij})), \tag{22}$$

where the standard deviation std_{ij} represents the score obtained by student i on the question containing knowledge concept k. From Table 3, it can be seen that although the AVG_{res} of each dataset is quite different, $STD_{res>1}$ is close. Therefore, we consider that the knowledge status of students in these data sets is stable and reasonable, and can be further tested.

5.2 Baseline

We compared CDMFKC with the following baselines on the same dataset:

- **Item Response Theory (IRT)** [23].
- **Multidimensional Item Response Theory (MIRT)** [25].
- **Deterministic Input Noise And Gate Model (DINA)** [13].
- **Probabilistic Matrix Factorization (PMF)** [17].
- **Neural Cognitive Diagnosis Model (NCDM)** [31].
- **Deep Knowledge Tracing (DKT)** [24].

All models are implemented by PyTorch using Python 3.7, all experiments are run on devices configured as CPU@3.00GHz, GPU RTX 1650Ti and 16GB RAM.

5.3 Analysis of Results

To verify the accuracy of student achievement prediction, our paper evaluates from the classification and regression aspects, including accuracy (Accuracy), root mean square error (RMSE), and area under curve (AUC).

Comparison of knowledge concept impact. We conduct experiments to compare the effect of h_j^{imp} without considering the training factors h_j^{disc} and h_j^{diff}, and g_j^{imp} including both on the prediction accuracy. The experimental results are shown in Table 4. We observe that the model with g_j^{imp} has higher prediction accuracy on the three datasets, which also demonstrates the effectiveness of our processing.

Prediction accuracy analysis. The students' skill proficiency vector is obtained through cognitive diagnosis, which reflects the students' proficiency in each knowledge concept. However, it is difficult to obtain the real knowledge status of students, which makes it difficult to evaluate the performance of cognitive diagnostic models. In practice, the results of cognitive diagnosis are usually obtained by predicting the performance of students, so we can use an indirect way to evaluate the performance of the model, which is to predict the performance of students in unfinished exercises [12].

Tables 5 and 6 summarize the experimental results of all models on student performance prediction tasks, with the best results shown in bold. All models are evaluated using 5-fold cross-validation, ± representing standard deviations. From Tables 5 and 6, we can observe that compared with NCDM on six datasets, CDMFKC predicted an improvement of about 3.5%, 2.6%, 3.4%, 2.6%, 1.1% and 3% respectively, indicating the validity of our model. On the other side, combined with the parameter comparisons in Table 2, we find that models with fewer considerations perform worse (such as DINA and PMF), which also proves that fully utilizing the information of the learning system can improve the accuracy of the model. The accuracy of DKT is higher on the bridge dataset, which may be because the dataset is more suitable for knowledge tracking.

Ablation experiments. To understand how the impact of knowledge concepts, guessing parameters, and slipping parameters affect the performance of the model, ablation experiments were

Table 5: Experimental results on predicting student grades on three real datasets. Math does not meet DKT requirements.

Model	ASSIST			Math1			Math2		
	ACC	RMSE	AUC	ACC	RMSE	AUC	ACC	RMSE	AUC
DINA [13]	0.650±.001	0.467± .001	0.676± .001	0.593±.001	0.487±.001	0.686±.001	0.592±.001	0.475±.002	0.683±.001
IRT [23]	0.674±.002	0.464± .002	0.685± .001	0.693±.002	0.478±.001	0.705±.001	0.700±.001	0.457±.001	0.703±.001
MIRT [25]	0.701±.001	0.461± .001	0.709± .002	0.719±.002	0.464±.001	0.733±.001	0.712±.002	0.448±.002	0.736±.002
PMF [17]	0.661±0.001	0.476± .001	0.732± .002	0.684±.001	0.467±.001	0.715±.001	0.696±.001	0.464±.001	0.734±.001
DKT [24]	0.718±0.001	0.442± .001	0.742± .001	-	-	-	-	-	-
NCDM [31]	0.719±.008	0.439± .002	0.749± .001	0.714±.002	0.439±.001	0.796±.001	0.698±.003	0.450±.002	0.771±.001
CDMFKC	**0.762±.001**	**0.431± .001**	**0.750± .001**	**0.740±.003**	**0.437±.001**	**0.797±.001**	**0.732±.002**	**0.442±.001**	**0.778±.002**

Table 6: Experimental results on predicting student grades on three real datasets.

Model	Algebra1			Algebra2			Bridge		
	ACC	RMSE	AUC	ACC	RMSE	AUC	ACC	RMSE	AUC
DINA [13]	0.623±.001	0.486±.001	0.646±.001	0.643±.001	0.477±.001	0.653±.001	0.652±.001	0.455±.002	0.673±.001
IRT [23]	0.764±.002	0.469±.002	0.775±.001	0.713±.001	0.453±.001	0.725±.001	0.760±.001	0.456±.001	0.743±.001
MIRT [25]	0.746±.001	0.454± .001	0.786± .001	0.733±.002	0.461±.001	0.743±.001	0.756±.001	0.455±.002	0.776±.001
PMF [17]	0.721±0.001	0.476± .001	0.727± .002	0.724±.001	0.445±.001	0.743±.001	0.747±.001	0.454±.001	0.765±.001
DKT [24]	0.789±0.001	0.385± .001	**0.816± .002**	0.798±.002	0.383±.001	**0.805±.001**	**0.854±.001**	0.371±.001	**0.801±.001**
NCDM [31]	0.800±.001	0.387± .002	0.779± .001	0.817±.002	0.366±.001	0.779±.001	0.818±.001	**0.364±.001**	0.783±.001
CDMFKC	**0.826±.002**	**0.378± .001**	0.794± .001	**0.828±.001**	**0.365±.001**	0.786±.001	0.847±.002	0.368±.001	0.778±.001

Table 7: Ablation experiment results. The horizontal lines in the first column represent the components we removed in our experiments. GS is the guessing parameter and slipping parameter, and FKC is the knowledge concept impact parameter.

Model	ASSIST			Math1			Math2		
	ACC	RMSE	AUC	ACC	RMSE	AUC	ACC	RMSE	AUC
GSCDMFKC	**0.770±.002**	**0.431± .001**	**0.751± .001**	**0.747±.002**	**0.436±.001**	**0.797±.001**	**0.739±.002**	**0.442±.002**	**0.780±.002**
~~GS~~**CDMFKC**	0.762±.001	0.431± .001	0.750± .001	0.740±.001	0.437±.001	0.797±.001	0.732±.002	0.442±.001	0.778±.002
GSCDM~~FKC~~	0.725±.001	0.439± .001	0.750± .002	0.717±.001	0.438±.001	0.799±.001	0.705±.001	0.443±.002	0.777±.001
~~GS~~**CDM**~~FKC~~	0.719±.008	0.439± .002	0.749± .001	0.714±.002	0.441±.001	0.797±.001	0.698±.001	0.450±.002	0.771±.001

Table 8: Ablation experiment results. The horizontal lines in the first column represent the components we removed in our experiments. GS is the guessing parameter and slipping parameter, and FKC is the knowledge concept impact parameter.

Model	Algebra1			Algebra2			Bridge		
	ACC	RMSE	AUC	ACC	RMSE	AUC	ACC	RMSE	AUC
GSCDMFKC	**0.832±.001**	**0.379± .001**	0.794± .001	**0.835±.002**	0.365±.001	**0.786±.001**	**0.858±.001**	0.370±.001	**0.778±.001**
~~GS~~**CDMFKC**	0.826±.002	0.378± .001	0.794± .001	0.828±.001	**0.365±.002**	0.785±.001	0.847±.002	0.368±.001	0.778±.001
GSCDM~~FKC~~	0.811±.001	0.372± .001	**0.813± .002**	0.812±.001	0.366±.001	0.779±.001	0.814±.001	**0.363±.001**	0.784±.001
~~GS~~**CDM**~~FKC~~	0.800±.001	0.387±.002	0.779±.001	0.817±.002	0.366±.001	0.779±.001	0.818±.001	0.364±.001	0.783±.001

performed on six datasets, where the letters with horizontal lines are the components removed.

Tables 7 and 8 show the results of the extended experiments. In these two tables, we observe that each optimization removed leads to a certain reduction in the accuracy of the model, which indicates that the optimizations proposed in this paper are all quite important.

Interpretable analysis. To demonstrate the interpretability of our diagnostic model, we proceed with some experiments.

Intuitively, students' mastery of knowledge concepts is proportional to their probability of answering correct questions. That is, for knowledge concept k, if student a has a higher level of mastery than student b, it is easier for student a to correctly answer questions related to knowledge concept k than student b. We adopt Degree

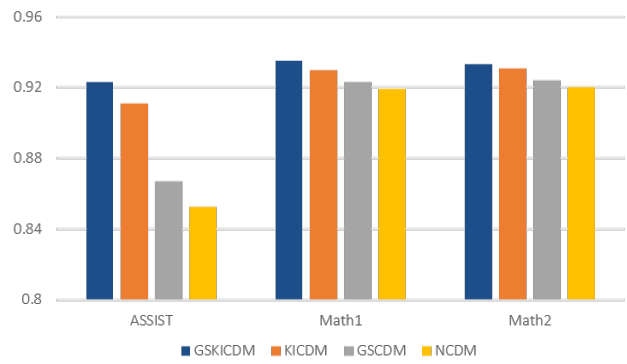

Figure 3: DOA results of model interpretability on ASSIST , Math1 and Math2.

of Agreement (DOA) [12] as the evaluation metric for model interpretability. For knowledge concept k, $DOA(k)$ can be expressed as:

$$\frac{1}{P}\sum_{x=1}^{N}\sum_{y=1}^{N}\delta(V_{xk},V_{yk})\frac{I_{jk}\wedge J(j,x,y)\wedge\delta(r_{xj},r_{yj})}{I_{jk}\wedge J(j,x,y)}, \tag{23}$$

where $P = \sum_{x=1}^{N}\sum_{y=1}^{V}\delta(V_{xk},N_{yk})$. $\delta(x,y) = 1$ if $x > y$ and $\delta(x,y) = 0$ otherwise. V_{xk} indicates the proficiency of student x in knowledge concept k. $I_{jk} = 1$ if exercise j examines knowledge concept k, otherwise $I_{jk} = 0$. $J(j,x,y) = 1$ if both students x and y have done exercise j, else $J(j,x,y) = 0$. r_{xj} represents the score of student x on question j. The DOA of the model is obtained by averaging the DOA of all knowledge concepts.

We compare the DOA value with NCDM because there is no clear correspondence between the potential features and knowledge concepts of other models. We selected ASSIST, Math1 and Math2 for experiments. The results shown in Figure 4 show that GSCDMFKC is superior to all other baselines in model interpretability. We also observed that the DOA value of NCDM was lower than that of CDMFKC, indicating that the model considering the impact of knowledge concepts is more interpretable. Moreover, the DOA value of CDMFKC is lower than that of GSCDMFKC, which also proves that considering guessing and slipping parameters can improve the interpretability of the model.

Case Analysis. To demonstrate our diagnostic results, we selected a student's CDMFKC diagnostic results on the Math1 dataset. As shown in Figure 4, the table in the upper part of the figure shows Q matrix for the four questions and the students' actual responses. The bar graph in the lower part of the graph represents the students' familiarity with each knowledge concept, and the line graph represents the difficulty of the corresponding knowledge concept of the test question. We can observe from the graph that students are more likely to respond correctly when their proficiency meets the practice requirements. For example, Exercise 3 requires students to master "Spatial imagination", and the students' proficiency in this knowledge concept is 0.473, while the corresponding difficulty of Exercise 3 is 0.432. The proficiency level of the student is higher than required, so the student is diagnosed with the correct answer.

	Trigonometric Function	Logarithm versus Exponential	Reasoning and Demonstration	Calculation	Response
Exercise 1	1	1	0	0	√
Exercise 2	1	0	0	1	√
Exercise 3	1	0	0	1	×
Exercise 4	1	0	1	1	×

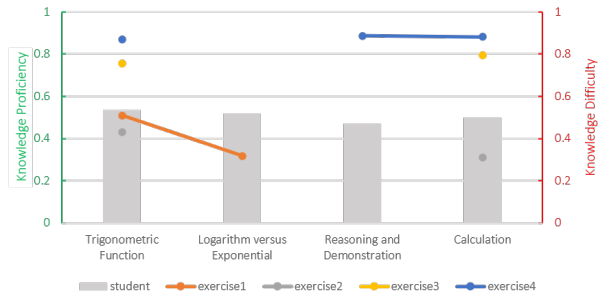

Figure 4: Diagnostic example of a student in Math1. The upper part is the Q-matrix of 4 exercises and the corresponding answer records. The bottom half shows the knowledge proficiency of the diagnosed student and the conceptual difficulty of each exercise.

6 CONCLUSION

This paper proposes a new neural network-based cognitive diagnosis model CDMFKC for cognitive diagnosis of students. Specifically, we use neural networks to obtain rich information in the learning system, introduce the impact of knowledge concepts and correlate them with exercise discrimination and knowledge difficulty, and dynamically design judgment boundaries to improve the model. On this basis, the GSCDMFKC model was obtained by further designing guessing parameters and slipping parameters. Experimental results on real data sets show that our model is more accurate and interpretable. We also performed ablation experiments to demonstrate that our optimization is reasonable and effective. The neural network used in this paper is relatively simple. It is interesting to use more complex neural network as the basis of cognitive diagnosis model.

ACKNOWLEDGMENTS

This paper was supported by National Natural Science Foundation of China (Nos. 62077028, 61906216 and 61877029), the Guangdong Basic and Applied Basic Research Foundation (Nos. 2021A1515011873, 2021B0101420003, 2021B1515120048 and 2020A1515010642), the Specially Creative University Project for Guangdong (Nos. 2019KTSCX010), the Science and Technology Planning Project of Guangzhou (Nos. 202206030007, 202102080307 and 201902010041), the Teaching Reform Research Projects of Jinan University (No. JG2021112), the Project of Guangxi Key Laboratory of Trusted Software (No. kx202007), and the High Performance Public Computing Service Platform of Jinan University.

REFERENCES

[1] Patricia A Alexander and Gavin TL Brown. 2017. *Assessment of Student Achievement.* Routledge.

[2] Arpan Basu, Avishek Garain, and Sudip Kumar Naskar. 2019. Word Difficulty Prediction Using Convolutional Neural Networks. In *TENCON 2019-2019 IEEE Region 10 Conference (TENCON)*. IEEE, 1109–1112.

[3] Lijia Chen, Pingping Chen, and Zhijian Lin. 2020. Artificial Intelligence in Education: A Review. *IEEE Access* 8 (2020), 75264–75278.

[4] Yunxiao Chen, Xiaoou Li, Jingchen Liu, and Zhiliang Ying. 2018. Recommendation System for Adaptive Learning. *Applied Psychological Measurement* 42, 1 (2018), 24–41.

[5] Yuying Chen, Qi Liu, Zhenya Huang, Le Wu, Enhong Chen, Runze Wu, Yu Su, and Guoping Hu. 2017. Tracking Knowledge Proficiency of Students with Educational Priors. In *Proceedings of the 2017 ACM Conference on Information and Knowledge Management*. 989–998.

[6] Jimmy De La Torre. 2009. DINA Model and Parameter Estimation: A Didactic. *Journal of Educational and Behavioral Statistics* 34, 1 (2009), 115–130.

[7] Jimmy De La Torre. 2011. The Generalized DINA Model Framework. *Psychometrika* 76, 2 (2011), 179–199.

[8] Jimmy De La Torre and Jeffrey A Douglas. 2004. Higher-Order Latent Trait Models for Cognitive Diagnosis. *Psychometrika* 69, 3 (2004), 333–353.

[9] Jimmy De La Torre and Jeffrey A Douglas. 2008. Model Evaluation and Multiple Strategies in Cognitive Diagnosis: An Analysis of Fraction Subtraction Data. *Psychometrika* 73, 4 (2008), 595–624.

[10] Louis V DiBello, Louis A Roussos, and William Stout. 2006. 31a Review of Cognitively Diagnostic Assessment and A Summary of Psychometric Models. *Handbook of Statistics* 26 (2006), 979–1030.

[11] Guillaume Durand, Nabil Belacel, and Cyril Goutte. 2015. Evaluation of Expert-based Q-matrices Predictive Quality in Matrix Factorization Models. In *European Conference on Technology Enhanced Learning*. Springer, 56–69.

[12] Francois Fouss, Alain Pirotte, Jean-Michel Renders, and Marco Saerens. 2007. Random-Walk Computation of Similarities between Nodes of A Graph with Application to Collaborative Recommendation. *IEEE Transactions on knowledge and Data Engineering* 19, 3 (2007), 355–369.

[13] Edward H Haertel. 1989. Using Restricted Latent Class Models to Map The Skill Structure of Achievement Items. *Journal of Educational Measurement* 26, 4 (1989), 301–321.

[14] Zhenya Huang, Qi Liu, Enhong Chen, Hongke Zhao, Mingyong Gao, Si Wei, Yu Su, and Guoping Hu. 2017. Question Difficulty Prediction for Reading Problems in Standard Tests. In *Proceedings of the Thirty-First AAAI Conference on Artificial Intelligence*, Vol. 31. 1352–1359.

[15] Yasuo Kinouchi and Kenneth James Mackin. 2018. A Basic Architecture of An Autonomous Adaptive System with Conscious-Like Function for A Humanoid Robot. *Frontiers in Robotics and AI* 5 (2018), 30.

[16] Qiuqiang Kong, Yin Cao, Turab Iqbal, Yuxuan Wang, Wenwu Wang, and Mark D Plumbley. 2020. Panns: Large-Scale Pretrained Audio Neural Networks for Audio Pattern Recognition. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 28 (2020), 2880–2894.

[17] Yehuda Koren, Robert Bell, and Chris Volinsky. 2009. Matrix Factorization Techniques for Recommender Systems. *Computer* 42, 8 (2009), 30–37.

[18] Jacqueline Leighton and Mark Gierl. 2007. *Cognitive Diagnostic Assessment for Education: Theory and Applications*. Cambridge University Press.

[19] Qingbiao Li, Fernando Gama, Alejandro Ribeiro, and Amanda Prorok. 2020. Graph Neural Networks for Decentralized Multi-Robot Path Planning. In *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 11785–11792.

[20] Qi Liu, Zhenya Huang, Yu Yin, Enhong Chen, Hui Xiong, Yu Su, and Guoping Hu. 2019. Ekt: Exercise-Aware Knowledge Tracing for Student Performance Prediction. *IEEE Transactions on Knowledge and Data Engineering* 33, 1 (2019), 100–115.

[21] Qi Liu, Shiwei Tong, Chuanren Liu, Hongke Zhao, Enhong Chen, Haiping Ma, and Shijin Wang. 2019. Exploiting Cognitive Structure for Adaptive Learning. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. 627–635.

[22] Qi Liu, Runze Wu, Enhong Chen, Guandong Xu, Yu Su, Zhigang Chen, and Guoping Hu. 2018. Fuzzy Cognitive Diagnosis for Modelling Examinee Performance. *ACM Transactions on Intelligent Systems and Technology (TIST)* 9, 4 (2018), 1–26.

[23] Frederic Lord. 1952. A Theory of Test Scores. *Psychometric Monographs* (1952).

[24] Chris Piech, Jonathan Bassen, Jonathan Huang, Surya Ganguli, Mehran Sahami, Leonidas Guibas, and Jascha Sohl-Dickstein. 2015. Deep Knowledge Tracing. *Proceedings of the 28th International Conference on Neural Information Processing Systems* 1 (2015), 505–513.

[25] Mark D Reckase. 2009. Multidimensional Item Response Theory Models. In *Multidimensional Item Response Theory*. Springer, 79–112.

[26] Ritter S Gordon G.J Koedinger K.R Stamper J, Niculescu-Mizil A. 2010. Data Set from KDD Cup 2010 Educational Data Mining Challenge. http://pslcdatashop.web.cmu.edu/KDDCup/downloads.jsp.

[27] Xueying Tang, Yunxiao Chen, Xiaoou Li, Jingchen Liu, and Zhiliang Ying. 2019. A Reinforcement Learning Approach to Personalized Learning Recommendation Systems. *Brit. J. Math. Statist. Psych.* 72, 1 (2019), 108–135.

[28] Nguyen Thai-Nghe and Lars Schmidt-Thieme. 2015. Multi-Relational Factorization Models for Student Modeling in Intelligent Tutoring Systems. In *2015 Seventh International Conference on Knowledge and Systems Engineering (KSE)*. IEEE, 61–66.

[29] Andreas Toscher and Michael Jahrer. 2010. Collaborative Filtering Applied to Educational Data Mining. *KDD Cup* (2010).

[30] A Voulodimos, N Doulamis, A Doulamis, and E Protopapadakis. 2018. Deep Learning for Computer Vision: A Brief Review. *Computational Intelligence and Neuroscience* 2018 (2018), 7068349–7068349.

[31] Fei Wang, Qi Liu, Enhong Chen, Zhenya Huang, Yuying Chen, Yu Yin, Zai Huang, and Shijin Wang. 2020. Neural Cognitive Diagnosis for Intelligent Education Systems. In *Proceedings of the Thirty-Four AAAI Conference on Artificial Intelligence*, Vol. 34. 6153–6161.

[32] Runze Wu, Qi Liu, Yuping Liu, Enhong Chen, Yu Su, Zhigang Chen, and Guoping Hu. 2015. Cognitive Modelling for Predicting Examinee Performance. *Proceedings of the 24th International Conference on Artificial Intelligence*, 1017–1024.

[33] Xiaolu Xiong, Siyuan Zhao, Eric G Van Inwegen, and Joseph E Beck. 2016. Going Deeper with Deep Knowledge Tracing. *Proceedings of the 9th International Conference on Educational Data Mining* (2016), 545–550.